

Weatherization Tutoring for the Homeowner

The key to a happy client is not just work that's done well, but work that involves the client's participation in the project.

by Henderson Callender

Home invasions can be very distressful for families—even benign invasions, like visits by work crews out to perform weatherization services. Homeowners who contract to have their house weatherized certainly want the work to get done. Nevertheless, there is a certain amount of trepidation involved in letting strangers into your house accompanied by tools and a mission. You think: How disruptive is this going to be?

For the past 13 years, our nonprofit organization, Community Environmental Center (CEC), has performed weatherization services in metro New York City, “invading” some 200 one- to four-family homes each year for fee-for-service jobs as well as no-charge services under New York State’s Weatherization Assistance program (WAP) for low-income families (see “From Wall Street to Green Construction” and “About Community Environmental Center”).

Based on our experience, CEC has learned that the key to a happy client is not just work that’s done well, but work that involves the client’s participation in the project.

Making the Client Comfortable

Client participation happens through on-site education. Weatherization professionals explain to the client what the crew intends

From Wall Street to Green Construction

Richard M. Cherry is the founder, president, and chief executive officer of the Community Environmental Center (CEC). Early in his career, Cherry left a lucrative position as a Wall Street lawyer to devote himself to less-fortunate New Yorkers. He joined the New York Urban Coalition (NYUC), a not-for-profit project started by the corporate community to bring labor, businesses, and government together to deal with poverty and racial strife. Over a 20-year career there, Cherry rose to become executive vice president of the NYUC and president of the NYUC Housing Group, managing housing, community development, and employment and training programs and providing legal consultation and leadership to various citywide policy advocacy efforts.

When the Urban Coalition dissolved in 1994, Cherry used his broad network and knowledge of publicly supported housing and community development programs to create the CEC, the first New York City nonprofit to focus exclusively on environmental issues of housing and development.

Over the years since Cherry founded CEC, the organization has emerged as the largest not-for-profit energy conservation contractor in New York State, as well as an overall promulgator of green values in building construction.

to do and what each step should accomplish to help make the structure more weathertight and energy efficient. For many organizations performing weatherization services across the country, this may seem standard operating procedure. However, it is advice that should be reiterated. Routine makes for carelessness. And a slipup—something as simple as a worker who seems brusque—can sour a client/work crew relationship. We encourage our work crews to respect the house and to respect the client.

The result of our taking the time and trouble to explain each step we take is feedback from clients that is

99% affirmative and that often leads clients to refer us to their friends. As one Brooklyn, New York, homeowner told us, “I was a little scared [about the weatherization process]; this was an unknown quantity to me . . . But they told me all I needed to know about what they were going to do and why. When they said they would come at 9, they came at 9. They said it would take three or four days—it did.”

The first step in the weatherization process is most often a phone call to introduce ourselves to the client and present an overview of the weatherization process and what we hope to accomplish. We confirm an



Henderson Callender of the Community Environmental Center (CEC) points out where cracks around the top of the riser should be repaired as part of the air-sealing program.

About Community Environmental Center

Community Environmental Center (CEC) has been in the forefront of the growing effort to improve the performance of the built environment in New York, especially in the New York City metro area. As a not-for-profit, the firm is uniquely dedicated to this goal, which includes reducing environmental pollution, limiting the need for expensive public infrastructure projects, and assisting the lower-income population by reducing the cost of utilities.

High-Performance Building Services

CEC, established in 1994, focused initially on the energy conservation needs of existing residential buildings. With the advent of the U.S. Green Building Council's LEED system and increasing market interest in making buildings more energy and water efficient and healthy to live in, CEC added staff to provide green building and LEED certification consulting services. CEC's Engineering Services and Technical Services departments now have 15 professionals experienced in various aspects of high-performance building, including five engineers. Six members are LEED accredited. While the majority of CEC's projects involve existing residential buildings, our client base includes owners of commercial and industrial facilities and new construction of all types. Eight of CEC's projects are registered to be LEED certified at various levels. CEC has published *Affordably Green in NYC*, a guidebook to green design for low-income housing in New York City.

CEC's experience ranges from contracts to develop ambitious efficiency standards for an entire city agency and for a major New York State Energy Research and Development Agency (NYSERDA) housing program, to overseeing the work of other firms in implementing the standards and to consulting and contract work on individual buildings.

In a large urban area like New York City, the stock of existing buildings far outweighs the amount of new construction. CEC has played an important role in the upgrading of existing buildings as it continues to be the largest not-for-profit energy conservation contractor in the state, currently managing over \$8 million in annual contracts from the State Division of Housing and Community Renewal (DHCR), NYSERDA, and private building owners. CEC has reached more than 250,000 people with energy conservation programs that reduced their utility costs by an estimated \$22 million. This energy use reduction translates into 44,000 tons less of CO₂ being emitted into the air per year.

With its knowledge of the available federal, state, and local government programs, CEC is able to access applicable public incentives for its clients and guide them through the application and compliance procedures. CEC directs clients as appropriate to the private energy market programs in peak load reduction to create income incentives for energy efficiency.



Callender records data from a blower-door test indicating the degree of air infiltration from unconditioned space, such as attics and crawlspaces.

Environmental Education

CEC began its public environmental education effort in 2002, when it won a city contract to construct an environmental education center at Stuyvesant Cove Park in Manhattan. A year later, CEC created a related not-for-profit, Solar One, to manage and guide the project forward. Staff provide specialized environmental issues classes for students and adults and host various cultural events out of a small classroom structure powered largely by solar energy, which serves as a living example of solar power use. Designs are under way for Solar 2, an 8,000 ft² facility to house a larger program and to demonstrate the feasibility of green design. It will be a net zero energy building.

Deconstruction Service and Materials Outlet

CEC established its salvage material warehouse and deconstruction service, Build It Green!/NYC, in support of its goals to assist building owners in achieving LEED certification and to support affordable housing efforts in New York City. The service addresses two of the LEED criteria dealing with materials and resources and has diverted tons of reusable materials from New York City's waste stream. Builders and homeowners are able to buy high-quality used appliances, cabinets, doors, and other items from our Build It Green!/NYC warehouse.

Proud to Serve New York City

CEC's staff reflects its pledge to social equity, boasting a high percentage of women and various minorities in both management and operations. CEC has deep roots in the New York City community through its relationships with government agencies and the many community service organizations, housing authorities, and building managers and owners with whom it has worked. This, together with its unusual combination of experience in all kinds of buildings, from single-family homes to large complexes like Co-op City and commercial facilities, and its deconstruction and public education services, enables CEC to play an important role in transforming New York City into a sustainable high-performance city.

appointment and give the householder an estimate on how long the job will take. We don't just explain what we do; we emphasize the benefits that will come from weatherizing the home.

We begin the relationship like a doctor: We take a history of the "patient." How long have you lived there? Have you had construction done before? What kind of work? Do you have ongoing service contracts?

Energy Audit as a Teaching Tool

Following the initial contact, when we go to the home for the energy audit phase, it's time to do more explaining. We've done an overview of the dwelling from the outside; now we want to tour the house with the client to scope the work that must be done, while inspecting the heating system and activating detectors to check for gas leaks and CO buildup.

Meanwhile, we are also keeping an eye out for unsafe conditions, such as loose wiring, and water seeping into the basement. If we do spot a hazard, more often than not an electrical problem, we must be diplomatic. We explain to the client that correcting such a problem is not the responsibility of weatherization work crews. The client must redress the problem with qualified professionals, like electricians, before the work crew arrives.

Explaining the Work Scope

With the audit completed, we discuss with the client what we think needs to be done to the house—and this does not always accord with what the client thinks or expects should be done (see "The Use of Energy Labels in Two European Countries," *HE* Mar/Apr '08, p. 8). We steer the client in the right direction, again by explaining why a different approach to making the home weathertight may be a more cost-effective way.

There seems to be an assumption, for example, that "weatherization" means "new windows." This is not necessarily so. We might explain why a strategy like adding insulation around windows

would be more effective in retaining heat than replacing windows. We want clients to clearly understand what renovations they will get—and what they won't get—before the auditor leaves the home.

Make Sure the Client Is at Home

When setting a date and time to start work, it is important to stress that the client be on premises. There are times when a work crew arrives to find no one home. Or the homeowner decides at the last moment that he or she doesn't want the work done right now. Or the wife made an appointment with us, but the husband appears to say, "I don't want you to touch my house."

These kinds of situations are becoming increasingly rare, thanks to the public's growing awareness of environmental matters and the need many people feel to conserve energy. When we explain to clients the difference in energy use, and what that means in savings, between CFL bulbs and fixtures and conventional ones, they are much more appreciative than they might have been a few years ago.

Similarly, we get a more knowledgeable and appreciative response when we describe how the client will save on utility bills by replacing an old refrigerator (or dishwasher or clothes washer) with an Energy Star appliance, or how much more efficient a gas-driven drier is than an electric one. Never lose the opportunity to ballyhoo, and to demonstrate when possible, how a household's life will change, how much more comfortable the home will be, how the client will save dollars, with the upgrades being made to weatherize the home.

With the arrival of the work crew, we make another tour of the house, always in the company of the client. We will not work in the house if the client is not there, for various reasons. One reason is the security of private property, always a touchy subject for work crews, who must defend themselves if the client claims, for

instance, that an earring is missing from her dresser top.

Minimize Disruption

The work crew tries to work as much outside the bounds of the client's daily life as possible. For instance, insulation can often be blown in from outside the house. Otherwise, to install insulation and perform other disruptive tasks, we prepare clients in advance by asking them to remove clothing from closets, pull back carpeting, and get dogs and other pets out of the way. Work crews may have to move furniture from one side of the house to the other; they always do this carefully and in consultation with the client.

Since installing insulation under the floorboards in the attic often affects lights, we turn the lights on and leave them on after the job is completed to show the client that all is well.

When the job is completed, it is beneficial to point out various improvements, especially if they are not obvious to the eye. If the heating system has been replaced or revamped, we explain any new technology involved and how it works.

Safety Drill for Workers

An on-the-job education program also should target the crew, serving as a workday drill or constant-reminder program that emphasizes following safety guidelines. Patrick Goodluck, CEC's on-site manager, puts workers through their paces regarding proper use of protective clothing and equipment, including the so-called Tyvec space suits and face masks essential for such jobs as installing insulation in closed spaces, hard hats, and gloves. If a worker is being newly outfitted with protective clothing, does it fit properly? Is it comfortable?

This is established as our routine morning safety drill to check the proper working order of equipment, including extension ladders, triangular (step) ladders, extension cords, drills, saws, blowers, and vehicles. We have been trained in OSHA standards, and workers are made familiar with them.

weatherization



From left, Gregory Elrock and Keith Bernard take note of a client's assessment of what needs to be done to make his home weather tight.

For example, workers can't grow beards, so facemasks will fit snugly.

Among other procedures, the crew is drilled on the proper setup and securing of scaffolding; is taught that extended ladders are positioned at a 70° angle, with the base planted firmly 3 feet from the threshold or the wall; and is taught not to stand on the topmost platform of a triangular ladder.

When we start work on an older house, we assume that we will

encounter a certain amount of lead, and quite possibly asbestos. These materials may show up when we strip old insulation from pipes, for example, before performing blower tests. We watch for out-of-ordinary situations, such as water seepage, loose wires, and kerosene space heaters that must be removed. That's why we always wear protective clothing and masks for even the most routine tasks.

The goal is safety for the client and for the worker.

Sharing the Work

With the client comfortable with the work crew and briefed on everything that we are doing, and with work crew members always cognizant of the procedures that mean professionalism, we are accomplishing a sound job of weatherization for a household that now fully appreciates what goes into this kind of work—and what it accomplishes in making the home more comfortable, more energy

efficient, and more environmentally friendly.

This on-the-job tutoring of the client pays an added dividend: An educated client will take much better care of the house and be much more alert on ways to save energy. We've also recruited another client to the cause of improving the environment.

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